Week 8

Introduction to Academic Writing

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Check ins

- What experiences do you have with academic writing?
- How would you rate your own writing?
- What worries do you have for academic writing?

Introductions

- We'll cover introductions
- Literature reviews
- Next week, we'll cover methods
- Later this semester, we'll talk about structuring discussion sections

Finding Time to Write

- Pebble
  - 15-30 minutes
- Rock
  - 30 minutes to 1 hour
- Boulder
  - 1 hour+

What Goes Into Academic Writing

- Arguments
- Connections to past research
  - Citations
- RQs, hypotheses, purpose statement

What Does Not Go Into Academic Writing

- Reflections
- How you feel about the topic
- Making claims without evidence

Examples

- Students who feel supported do better in data science courses
- There is a need to further examine support systems for students in data science courses
- Students in data science courses use AI more than other disciplines
- Data science includes computing, statistics, and domain knowledge

Importance of Introductions

- Set the stage for your project
- Creates interest from audicnece
- Points out issues in the current literature
- Shows why the outcome you are interested in is a problem
- Places your study within a larger context

Academic Writing Is a Funnel (Big Picture –> Specifics)

- Talk about the overall problem
- What we know about the outcome/problem
- Talk about the literature focusing on the role of predictor(s) on outcome
- Your contribution to the literature + where there is disagreement in literaure
- Theory
- Purpose + RQs + Hypotheses

Overall Problem

- State what the problem is
  - Math readiness is bad in college
- Literature on math readiness
- The role of self-efficacy on math readiness
- Self-efficacy's relationship with math readiness looks different for students because of systematic issues (high school funding, high school teacher math competency)
  - Explore these differences by examining high schools across the United States from different income brackets
- Social Cognitive Theory: interactions among intrapersonal factors (self-efficacy), behaviors (math readiness), environment (funding, math teacher competency)
- Purpose

Citing

- You will fail this class if you only submit quotes for citations
- Paraphrase what you are referencing
- I will only allow 1 quote per paper, any more quotes = 0 on the paper
- Quoting is lazy writing

Primary & Secondary Sources

- **DO NOT** cite papers referenced from another paper
- A paper in the introduction citing another paper
  - **Read the original paper**
- Most authors barely read the article(s) they are citing
  - Authors misinterpret findings
  - They could paraphrase something differently

Introductions/Literature Reviews/Related Works are Field Dependent

- Association for Computing Machinery (ACM) conference proceedings
  - Introduction is composed of stating the problem
  - The purpose of your project
  - What you did in your project
  - The usefulness of your project
  - Think of it as a mini paper in your paper

Field Dependent Introduction

- The rest of the introduction focuses on literature
  - This will follow the rest of the slides
- **Base your paper on the literature you are reading**
  - My field combines introduction + literature review
  - Others have short introduction + more extensive literature reviews

First Paragraph in an Introduction

- Hook your audience into wanting to continue reading
  - Give your audience a number that shows them why your project is worth examining
  - "One in 12 incoming students tested below middle school proficiency in mathematics"
    - Secondary source to this article
    - From the primary source here
  - You don't need a thesis statement like in argumentative papers
    - Your purpose statement is the final paragraph in the introduction

Second Paragraph in Introduction

- Focus your audience's attention to justify the importance of the problem
  - Why this outcome needs to be further investigated/studied?
- Could include a systematic review on why your outcome needs to be further studied
  - Shows the extent of how much literature has covered your project topic
- End with a statement on why you are examining as predictors
- Ex: Math readiness is dropping by the time students get to college/university
  - Literature overall on math readiness in the US is bad
  - Math readiness is lower in elementary and high school which shows this starts early
  - That is why there is a need to explore students' math self-efficacy

Body Paragraphs in Introduction

- Reference the examination of a specific variable (inferential) OR literature on variables influence on the outcome (predictive analyses)
- Ex: Self-efficacy has been studied as a predictor of math readiness
  - Literature supporting/refuting/both
- With large amounts of literature on the topic –> focus on specifics that relate to your project
  - Self-efficacy in college students
- Transition to other variables/what is missing

Theory (if including)

- Find literature testing theory
- Add how your project is informed by your theory
- Cite your theory

Final Paragraph in Introduction

- "From these mixed findings, there is more a reason to focus on..."
- "Therefore, the purpose of this study is..."
- RQs
- Hypotheses

Writing

- Start by writing
  - Write anything
  - Put headers for sections to somewhat organize the draft
- Indicate where you'll put your citations
  - Paraphrase your citations
- Edit
- Edit
- Edit

JP's Writing Steps

- I read my literature + annotated bibliographies
- Either
  - Write a statement for each of my annotated bibliographies in a blank document
  - Start writing and then put CITE whenever I know I have something from my annotated bibliographies to include there
- Then fill in the blanks
- Reread sections and then change citations or go back to my annotated bibliographies to rewrite the statement to include in the introduction
- Write in order of:
  - Purpose statement
  - Opening paragraph
  - Second paragraph
  - Body paragraphs
  - Theory